

Title:- Food Trace cloud supply chain Tracker with QR provenance and Spoilage prediction.

Main use

Food Supply chain Manager:-

- * Tracks food products from farm to Consumer.
- * Takes action to reduce food waste and improve food quality.

Trace the Message Journey

1. User opens the food Trace mobile app.
2. User scans the QR code.
3. Mobile app sends an HTTPS request.
4. DNS resolves the server address.
5. TLS encrypts the connection.
6. Data travels using IP through the Internet.
7. Request reaches the API Gateway.
8. cloud server processes the request.
9. cloud Database retrieves product details.
10. Response is sent back through HTTP.

Cloud Services

Cloud Component

Mobile app

DNS

HTTPS/TLS

Purpose

User scans QR

find server

Secure communication

API Gateway	Receives requests
Authentication	Verifies user
Cloud Compute	Processes request
Database	Stores food records
Cloud Storage	Stores QR and
API Design	product data

API End point

POST /api/track-food

Method

Post

Input

QR code ID

User ID

Timestamp

Output

product name

Origin

Supply chain status

Expire date

Spoilage Prediction

Success Message

Security Decision

* Use HTTPS / TLS for Secure communication

* Authenticate users using JWT